



Evaluating the behavior and bond properties of FRP spike anchors under confined conditions and elevated temperature

Mehdi Khorasani^{a,b,*}, Giovanni Muciaccia^b, Andrea Nino Consiglio^b, Davood Mostofinejad^a

^a Department of Civil Engineering, Isfahan University of Technology (IUT), Isfahan 84156-83111, Iran

^b Department of Civil and Environmental Engineering, Politecnico di Milano, Milan 20133, Italy

ARTICLE INFO

Keywords:

FRP anchors
FRP Spike anchors
Strengthening
Confined conditions
Elevated temperature

ABSTRACT

The use of fiber-reinforced polymer (FRP) anchors in combination with the conventional externally bonded reinforcement (EBR) installation technique is an effective method to prevent or postpone the debonding of FRP sheets. However, their behavior under confined conditions and elevated temperatures is still unknown. In such a framework, in the first phase, the current research aims to assess pullout test results for single FRP anchors with dowels ranging from 50 to 75 mm in confined and unconfined conditions. To achieve this goal, three concrete slabs with the dimension of $1550 \times 1250 \times 250 \text{ mm}^3$ were cast. Experimental results showed that the bond strength of FRP anchors in confined conditions increased by about 50% compared to the unconfined conditions. Additionally, a comparison was made between the experimental results and the literature models. In the second phase, eleven pullout tests were conducted on the cylindrical specimens with the dimension of $150 \times 200 \text{ mm}^2$ (diameter $\times$ height) under confined conditions and elevated temperatures by considering different sustained load levels. The results showed as the temperature increases and FRP anchor is constantly loaded, it fails in a shorter time before the adhesive reaches its glass transition. Finally, a bond strength vs. temperature model was developed for FRP anchors in confined conditions and high temperatures.

1. Introduction

The combination of fiber-reinforced polymer (FRP) materials and externally bonded reinforcement (EBR) installation technique is an effective method for increasing the load-bearing capacity and/or ductility of reinforced concrete (RC) members [1–4]. Despite the benefits of EBR-FRP systems, such as their ease of application and rapid installation, the FRP composite sheets tend to debond prematurely from the concrete surface. Therefore, the high strength of carbon fiber-reinforced polymer (CFRP) materials is not able to be fully utilized to its capacity [5–7]. Nowadays, new methods for the installation of FRP materials like grooving methods [8–12] have been developed to prevent or delay premature debonding. Furthermore, for some FRP strengthening applications, such as shear strengthening of RC beams attached to a slab or T-shaped beams, a wrapping system may not be possible and External reinforcement is bonded only to the web. Therefore, researchers recommend U-shaped reinforcement. However, due to the location of cracks in the flange, the shear capacity of RC beams cannot be increased to its ultimate strength with U-shaped reinforcement [13].

In such cases, FRP anchors could provide a wrapping system for RC members. In fact, among different solutions, it has been demonstrated that FRP anchors are an effective solution in preventing or postponing premature debonding by ensuring the continuity of load paths from the FRP sheets into the structure or improving the FRP-to-concrete bond strength [14–16]. In fact, an FRP anchor increases the debonding strength of an external reinforcement by modifying the boundary conditions of the differential equation that governs its slip behavior [17]. The use of FRP anchors in reinforced concrete structures has been studied for strengthening or retrofitting a variety of elements, including beams [18,19], columns [20,21], beam-column connections [22–24], slabs [25,26], shear walls [27], and masonry elements [28,29].

FRP anchors can be installed using both wet and dry installation techniques. The wet procedure involves impregnating and rolling the fibers that have been cut to the desired size. Afterward, they are inserted into the hole filled with adhesive. In contrast, with the dry method, the rolled fibers soak into an adhesive-filled hole without being impregnated. As shown in Fig. 1, FRP anchors are made from fiber sheets or bundles of fibers from the same material used in FRP strips for strengthening procedures that are impregnated with epoxy adhesive as

* Corresponding author at: Department of Civil and Environmental Engineering, Politecnico di Milano, Italy and Department of Civil Engineering, Isfahan University of Technology (IUT), Iran.

E-mail addresses: mehdi.khorasani@mail.polimi.it, mehdi.khorasani@cv.iut.ac.ir, mehdi_kh11@yahoo.com (M. Khorasani).

<https://doi.org/10.1016/j.compstruct.2023.117407>

Received 15 February 2023; Received in revised form 18 July 2023; Accepted 26 July 2023

Available online 1 August 2023

0263-8223/© 2023 Elsevier Ltd. All rights reserved.

Nomenclature**Parameter Definition (unit)**

w_f	FRP strip width (mm)
h_{ef}	effective embedment length (mm)
d_a	diameter of anchor (mm)
d_h	diameter of anchor hole (mm)
TC1	thermocouple 1 temperature (°C)
TC2	thermocouple 2 temperature (°C)
U	unconfined conditions
C	confined conditions
α	fan angle (degree)
β	dowel angle (degree)
$f_{c,cube}$	compressive strength of cubic concrete (MPa)
ΔT_{oven}	temperature loading rate (°C/min)
R	ratio of sustained load (%)
N_{sust}	constant sustained load prior to temperature increase (kN)
τ_{sust}	bond stress of sustained load prior to temperature increase (MPa)
$F_{u,exp}$	experimental pullout capacity of FRP anchor (kN)
$F_{ua,exp}$	mean value of experimental pullout capacities of FRP anchors with the same conditions (kN)
Std. Dev. of $F_{u,exp}$	Standard deviation of experimental pullout capacity of FRP anchor

τ_u	bond strength along FRP anchor (MPa)
τ_{ua}	mean value of bond strength of FRP anchors with the same conditions (MPa)
Std. Dev. of τ_u	Standard deviation of bond strength along FRP anchor
t1	test duration (sec)
δ_f	the displacement at failure (mm)
T_f	weighted average of the temperature TC1 and TC2 at failure (°C)
T_{oven}	oven temperature (°C)
T_g	glass transition temperature (°C)
$F_{cone, Fuchs}$	concrete cone capacity according to Fuchs et al. (N)
$F_{cone, ACI 349-85}$	concrete cone capacity based on ACI 349-85 (N)
$F_{cone, Elig et al.}$	concrete cone capacity according to Eligehausen et al. (N)
τ	bond strength of FRP anchors according to uniform bond method (MPa)
τ_{avg}	mean bond strength of FRP anchors at confined conditions (MPa)
$F_{bond, Cook and Konz}$	pullout capacity of FRP anchor according to Cook and Konz (N)
$F_{bond, CEB}$	pullout capacity of FRP anchor based on CEB (N)
$F_{bond, current study}$	pullout capacity of FRP anchor according to current study (N)

an adhesive followed by one end inserted into a predrilled hole of a structure member, and the other end is bonded to the FRP sheet. Compared to anchors made of materials like metal, the FRP anchor fabrication process incorporates FRP strengthening strips made of the same material. This enhances the integrity of the strengthening system, thereby improving the performance of the entire system. In addition, FRP anchors are corrosion-resistant and flexible, making them suitable for wide-shaped FRP-strengthened members [30].

FRP anchors can be classified into two broad categories depending

on their dowel angle in relation to the reinforced member and the FRP sheet: straight anchors and bent anchors (Fig. 2). Basically, straight anchors transmit force from one member or part of it to another element. Consequently, these anchors are installed at the end of the FRP sheet, so that they align with the strengthening FRP sheet. In this case, the FRP anchor is aligned with the load, causing it to be subjected to pullout forces. On the other hand, bent anchors can be installed both alongside the FRP sheet as well as at its end. As long as they are placed at the end of the sheet, their function is to transfer force to the surrounding elements.

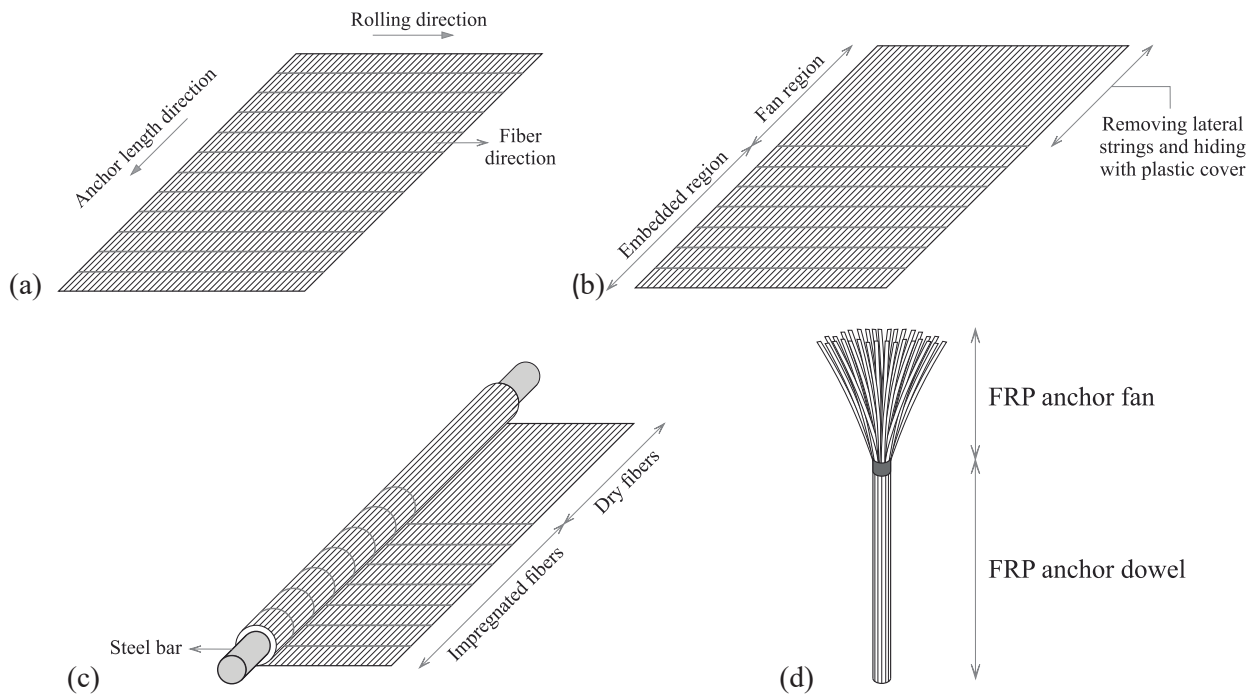


Fig. 1. The process of construction of FRP anchor: (a) cutting FRP sheet in the desired dimension, (b) removing lateral string in fan region, (c) impregnating both sides of FRP sheet in dowel region and rolling by a slender steel bar, (d) removing steel bar and opening fibers in fan region [31].

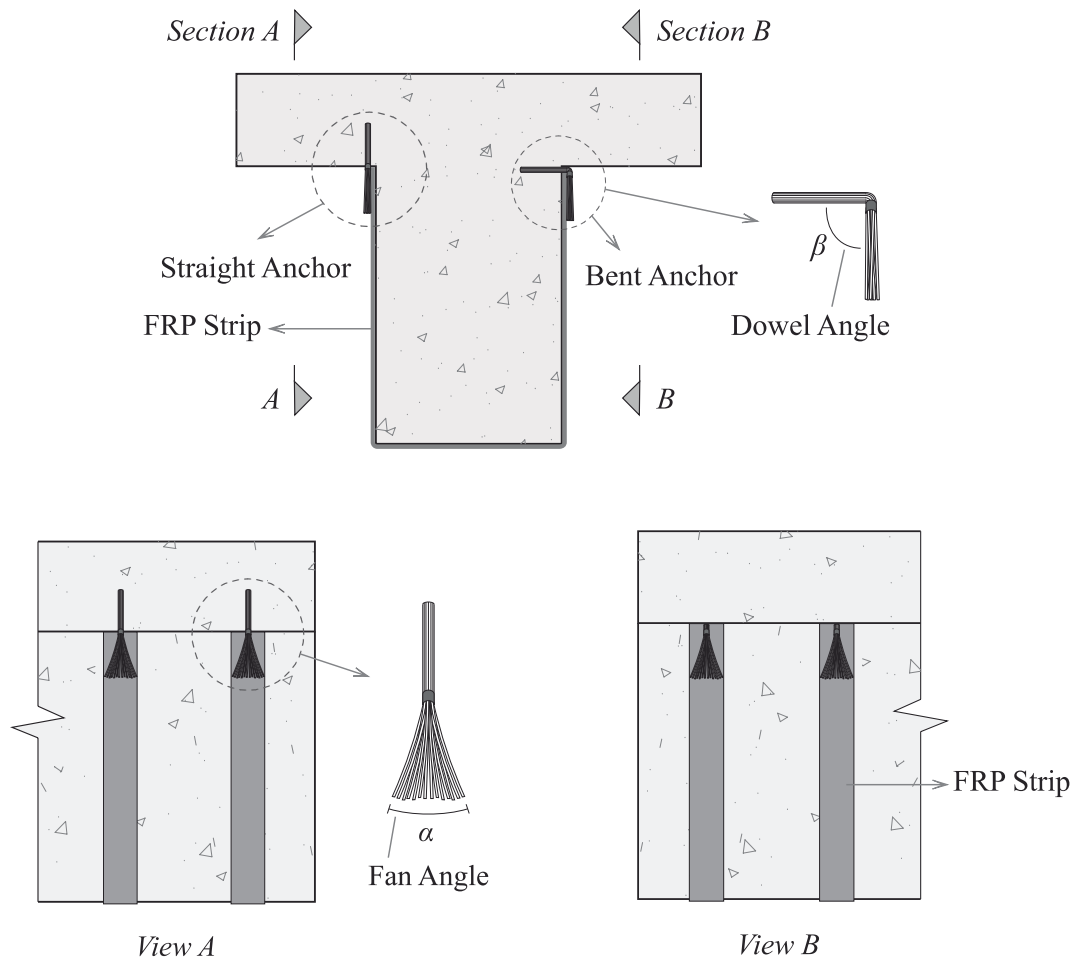


Fig. 2. Different types of FRP anchors [32].

Otherwise, they serve to prevent premature debonding. In this case, ideally, they would be oriented approximately perpendicular to the sheet at specific intervals. Furthermore, a bent anchor will be subjected to shear forces more than pullout forces.

There are many parameters that could affect the performance of FRP anchors. Anchor preparation and installation method [31,33,34], hole characteristics [35–38], anchor dowel angle with respect to load direction [16,33,39,40], embedment length (dowel length, h_{ef}) [41–45], amount of FRP used in FRP anchor [38,42,45–48], anchor patch geometry [38,49–52], fan geometry [38,43,53–56], spacing and configuration of FRP anchors [38,57] are the main parameters in the literature that their effect on the behavior of FRP anchors have been noticed by researchers. A comprehensive review of the behavior and the existing design procedures for FRP anchors can be found in Muciaccia et al. [32]. Moreover, del Rey Castillo et al. [16] examined the failure modes, design models, and strength of FRP anchors in conjunction with FRP-EBR systems to repair and/or strengthen RC members. Despite all the efforts to investigate the effect of various parameters on the behavior of FRP anchors, the effect of some parameters such as confinement, their behavior on the cracked concrete, their performance under elevated temperature or subjected to cyclic loadings like cyclic shear, cyclic tension actions and fatigue loads remains unknown. For instance, as FRP anchors are often used for retrofit applications, the concrete will likely be cracked, so the expected performance in cracked concrete should be investigated as post-installed anchors [58,59]. Additionally, a characteristic feature of earthquakes is cyclic loads, which are applied to the members of the structures and anchorages [60,61]. Among the research gaps regarding the behavior of FRP anchors, the current study attempts

to address the effect of confinement and elevated temperature on their bond resistance characteristics. In order to strengthen RC columns confined with internal lateral stirrups or external FRP reinforcement under axial load, confinement enhances the compressive strength of concrete [62,63]. Consequently, the behavior of an FRP anchor installed in confined sections could affect its behavior, strength, and failure modes. Appropriate placement of transverse reinforcement achieves confinement within a reinforced concrete member. However, according to ACI 355.4-19 [64] and EAD 330499 [65], confined testing can be conducted to determine bond strength to act as a reference when establishing anchor robustness with respect to e.g. temperature and creep conditions, as well as to establish a reference value to be used in the design process [66,67].

Fire safety is a crucial design criterion for utilizing FRP composites in buildings, parking structures, and infrastructure applications. However, there is currently no information available on the behavior of FRP anchors in the fire. In real-life applications, the reinforced members are protected by fire insulation to maintain a sufficiently low temperature on the unexposed surface. Consequently, structural members will be able to withstand the applied loads until a specific time under elevated temperatures without failing or exhibiting excessive deformations [68]. On the other hand, fire drastically reduces the stiffness and strength of FRP-strengthened reinforced concrete (RC) members [68–70]. Accordingly, applying FRP anchors could postpone the bond degradation at the FRP-concrete interface under elevated temperatures. In addition, the proper design of bonded anchors requires the prediction of long-term capacity, such as creep behavior. Typically, creep tests are conducted for at least three to six months at a given percentage of anchor capacity.

However, by utilizing a temperature test, the testing time could be reduced and an acceptable displacement-based failure criteria could be provided [71]. Therefore, it would be useful to understand how FRP anchors behave at elevated temperatures in order to properly design them. In literature, numerous temperature tests have been carried out on fasteners or post-installed steel anchors. Specifically, research on post-installed rebars showed that elevated temperatures cause a significant decrease in bond capacity [72–74]. Hence, the assessment of the effects of temperature on the bond properties of FRP anchors is necessary.

The present study deals with the determination of FRP anchors' strength in confined and unconfined conditions in order to evaluate the effect of confinement on the capacity of FRP anchors, as well as to use it as a reference in temperature tests. Then, to find out which models are most suitable for FRP anchors in the range of dowel lengths examined in this study, the experimental results are compared to classic strength models of post-installed anchors or fasteners. Finally, the behavior of FRP anchors under elevated temperatures is evaluated. Moreover, a model to predict the behavior of FRP anchors under elevated temperatures is presented.

2. Methodology and experimental program

2.1. Materials and their properties

Unidirectional CFRP sheets with a nominal design thickness of 0.17 mm were used to prepare the FRP anchors. The epoxy adhesive of Sikadur-330 was employed as the matrix phase of the CFRP composites. At room ambient temperature (23 °C) and according to the producer's datasheet, the adhesive of the strengthened specimens was cured for 7 days before testing. The properties of the fibers [75] and the adhesive [76] according to the manufacturer's data sheet are presented in Table 1. The Sikadur-330 adhesive consists of two parts (parts A and B (that are mixed in a ratio of 1:4. Additionally, the epoxy-impregnated fibers had a total thickness of 0.9 mm after adding adhesive to the dry fibers. It is worth noticing that the presented fiber's ultimate strength of 4000 MPa cannot be achieved using wet lay-up processes for FRP anchor preparation. In order to determine the ultimate capacity of the utilized FRP composite, on the basis of the ASTM D3039-00 [77] standard, the three CFRP coupons with dimensions of 50 × 250 mm (width × length) were prepared. According to uniaxial tension tests conducted on prepared coupons, carbon fibers impregnated with epoxy adhesives through wet layup had an average ultimate stress of 1990 MPa.

For testing FRP anchors at ambient tests (confined and unconfined conditions), FRP anchors were installed in reinforced concrete slabs optimized to prevent splitting failure induced by anchor loading. Three concrete slabs with the dimensions of 1550 × 1250 × 250 mm³ were cast to investigate the performance of the FRP anchors. On the other hand, for testing FRP anchors at elevated temperatures, cylindrical concrete specimens with dimensions of 150 mm diameter and 250 mm height were used. The concrete mix was designed for C20/25 class according to Eurocode 2 [78] by adopting round aggregate and CEM II 32.5 with some limestone additions. The mixed components are reported in

Table 1

Mechanical properties of materials used according to the manufacturer's datasheet.

Material	Type	Modulus of Elasticity in tension (GPa)	Ultimate tensile strength (MPa)	Thickness (mm)	Ultimate tensile strain (%)
Fibers	Sika Wrap-300C [75]	230	4000	0.17	1.7
Adhesive	Sikadur-330 [76]	4.5	30	–	0.9

Table 2. After casting, specimens were stored at room ambient temperature by avoiding direct sunlight exposure and, at curing onset, by keeping the cast surface wet. Specimens were stored at the same temperature conditions before testing. Cubes of side 150 mm were cast to determine the concrete compressive strength evaluated according to UNI EN 12390-3 [79]. The average compressive strengths of concrete cubes for each slab, $f_{c, \text{cube}}$, measured by compressive testing on the test day are given in Tables 3 and 4.

2.2. Preparing FRP anchors

In order to fabricate the FRP anchors, a rectangular piece of the FRP was cut to the desired size. After applying epoxy to both sides of the FRP piece, the anchors were formed by rolling the sheet into a cylinder. It should be noticed that before rolling the FRP, epoxy was applied to ensure full anchor impregnation (Figs. 3a to 4e). FRP Anchors made from 150 mm wide FRP sheets (w_f) that were rolled with an average diameter of 16 mm (d_a) after being impregnated and dried. The load was applied to FRP anchors using threaded rods with a diameter of 32 mm and a length of 100 mm, with a hole drilled through the length with a diameter of 20 mm (d_h). The rolled FRP sheet was then inserted into the threaded rod's hole (Fig. 3f). In the primary tests, the created hole had a smooth and prismatic section. Nevertheless, in the following tests, the internal face in the threaded rod was threaded separately after the bond failure in the sleeve was observed. In fact, when a hole is drilled within the threaded rod, the inside surface is smooth and does not bond properly to the anchor. Thus, the rod's internal surface was threaded similarly to its external surface.

2.3. Installation of FRP anchors

For investigating how confinement affects the performance of FRP anchors, they were installed in reinforced concrete slabs (Fig. 4a) optimized to minimize splitting failure. Drilling the hole at the prescribed diameter (20 mm) and embedment length (50 or 75 mm) was the first step in the installation process. Therefore, the adhesive was filled into the holes prior to inserting the anchor. In order to examine the behavior of FRP anchors under elevated temperature, concrete cylinders of 150 mm diameter and 200 mm height, encased in a 2 mm thickness of steel tube were used (Fig. 4b). The steel tube was used to create a mold for the concrete and to confine it, preventing premature failure. Additionally, it helped to distribute the temperature uniformly across the sample's surface. The drilled hole had a nominal depth of 50 mm with a diameter of 20 mm. For temperature acquisition, two thermocouples (Fig. 5) were placed along the embedded part in accordance with:

- TC1 placed at the deepest part of the FRP anchor.
- TC2 at 10 mm from the concrete surface.

Finally, the displacements at the FRP anchor-loaded end were monitored using two LVDTs.

2.4. Test setup

Two different test setups were adopted for confined and unconfined configurations. In confined tests, the reaction frame was equipped with a square steel plate at the bottom to provide confinement (Fig. 6b). In such cases, the confining plate was 35 mm in diameter and 200 mm in width. Conversely, in unconfined tests, the distance between the supports was 635 mm (Fig. 6a) to allow potential full concrete cone propagation. Testing was conducted using closed-loop servo-hydraulic machines with a load capacity of 100 kN for unconfined tests and 300 kN for confined tests. The test was conducted under displacement control at a displacement rate of 0.02 mm/s. Two 100 mm LVDTs were positioned symmetrically on both sides of the loading rod to monitor the slip at the loaded end. An HBM Spider8 data acquisition system was used to

Table 2

Mix design for concrete specimen.

Concrete class	Cement CEM II/B LL 32,5 R (kg/m ³)	Aggregate (kg/m ³)			Superplasticizer (% on cement mass)	Water (l/m ³)	Water/cement
		Fine (0–4)	Fine (0–8)	Coarse (6–16)			
C20/25	300	143	815	851	0.75	182	0.60

Table 3

Specimens' descriptions at ambient temperature.

No.	Specimen label	Concrete compressive strength, $f_{c,cube}$ (MPa)	Actual embedment depth, h_{ef} (mm)	Concrete confinement condition
1	L50-U-1	31.0	51	Unconfined
2	L50-U-2	27.3	50	Unconfined
3	L50-U-3	27.3	52	Unconfined
4	L75-U-1	27.3	75	Unconfined
5	L75-U-2	27.3	77	Unconfined
6	L75-U-3	27.3	76	Unconfined
7	L50-C-1	28.8	54	confined
8	L50-C-2	28.8	52	confined
9	L50-C-3	28.8	54	confined
10	L75-C-1	28.8	75.5	confined
11	L75-C-2	28.8	77	confined
12	L75-C-3	28.8	75	confined

* For all specimens, preparing FRP anchor and inserting it into sleeve and concrete were done simultaneously.

Table 4

Specimens' descriptions at elevated temperature.

No.*	Specimen label	$f_{c,cube}$ (MPa)	h_{ef} (mm)	N_{sust} (kN)	R (%)
13	L50-R10%-1	35.7	49.5	5	10
14	L50-R20%-1	35.7	50.6	9	20
15	L50-R20%-2	35.7	50.3	9	20
16	L50-R40%-1	35.7	50.1	18	40
17	L50-R40%-2	35.7	51.3	18	40
18	L50-R60%-1	35.7	50.2	27	60
19	L50-R60%-2	35.7	50.5	27	60
20	L50-R80%-1	35.7	50.8	36	80
21	L50-R80%-2	35.7	49.8	36	80

* For all specimens, preparing FRP anchor and inserting it into sleeve and concrete were done simultaneously.

acquire all data.

As for tests at elevated temperatures, the test setup consists of an electric oven with internal dimensions of $1000 \times 500 \times 500$ mm³, able to reach a maximum air temperature of 800 °C. The oven is mounted on a steel frame with a hydraulic jack attached. The reinforcing bar passes through a hole in the base of the oven and is connected to the jack by means of a special device. Layers of insulating material and steel plates are interposed between the loaded end of the specimen and the bottom of the oven that ensure both a proper load transfer and adequate insulation. Fig. 7 illustrates the test setup schematically. During the test, the slip of the FRP anchor at the loaded end is measured and automatically acquired at a frequency of 5 Hz.

2.5. Temperature loading protocol

The testing procedure considers the application of a constant load prior to temperature increase. This procedure is preferred over others because it stabilizes the temperature prior to increasing the load. In this way, the effects of the creep of the bonding agent and the initial mechanical damage (due to loading) can be considered [80]. These effects are relevant for structures subjected to fire or elevated temperatures during service. The initial load is applied at room temperature at a speed

of 0.30 kN/s until the desired target load level is reached. The target load was selected corresponding to 10%, 20%, 40%, 60%, and 80% of the pull-out capacity of FRP anchor obtained from the carried out tests in confined conditions and cold state (room temperature). The temperature is increased successively at an air temperature rate (ΔT_{oven}) of 10 °C/min while the load remains constant until the slip of the FRP anchor is detected. Fig. 8 reports the typical temperature and load evolution during a test. A quite uniform temperature distribution can be observed along the embedment depth so that such temperature can be associated with the temperature at the failure of the FRP anchor for that load level.

2.6. Test parameters and test code description

Test parameters and relevant codes for all tests at ambient temperature are presented in Table 3. The test code is composed as follows: LXX-Y-n, where XX denotes the actual embedment depth (or anchor length), Y indicates the boundary conditions ('C' for confined and 'U' for unconfined) and n indicates the repetition number. For all samples, the width of the FRP strip (w_f) for making the FRP anchors, the average diameter of the anchor after rolling (d_a), and the diameter of the hole (d_h) were 150 mm, 16 mm, and 20 mm, respectively.

For tests at elevated temperatures, anchors with an embedment length of 50 mm were selected. The load ratios of the constant load applied prior to temperature increase (10%, 20%, 40%, 60%, and 80%) represent a portion of the average capacity detected in the corresponding test series (45 kN). Code for tests at elevated temperatures is composed as LXX- RYY-n, where XX refers to the anchor length, YY indicates the load ratio, and n is the repetition number, as reported in Table 4. A 150 mm wide FRP strip was used to make the anchor. For all specimens, the diameter of the anchor after rolling and the anchor hole were 16 mm and 20 mm, respectively. Moreover, for all specimens at elevated temperatures, the compressive strength of cubic concrete samples was 35.7 MPa. The air temperature increases at a rate of 10 °C/min, followed by a constant load until the FRP anchor slips.

3. Results and discussion

3.1. Pull-out capacity and bond strength

In this section, the results of the experimental campaign are presented. In order to categorize and discuss the output of experimental work, the results related to conducting tests at ambient temperature and elevated temperatures have been distinguished from each other.

3.1.1. Ambient temperature

Load–displacement diagram of FRP anchors under confined and unconfined conditions at the ambient temperature is presented in Fig. 9. In order to avoid crowding the figure and make comparisons easier, one sample is taken from each group of specimens with similar characteristics. Based on the load–displacement diagrams for the samples at ambient temperature, the initial load path is quite linear, until the peak load is reached ($F_{u,exp}$). In confined specimens, FRP anchors started to pull out at this point. In this case, the adhesive and the FRP anchor completely separated, causing a pullout failure (Fig. 13). After peak load and failure occurrence, the load decreases with a low slope. The linear trend can be determined after the peak load has been reached. In this phase, the slip of the FRP anchor increases. In unconfined specimens, as FRP anchors reach their maximum capacity, cracks develop on concrete

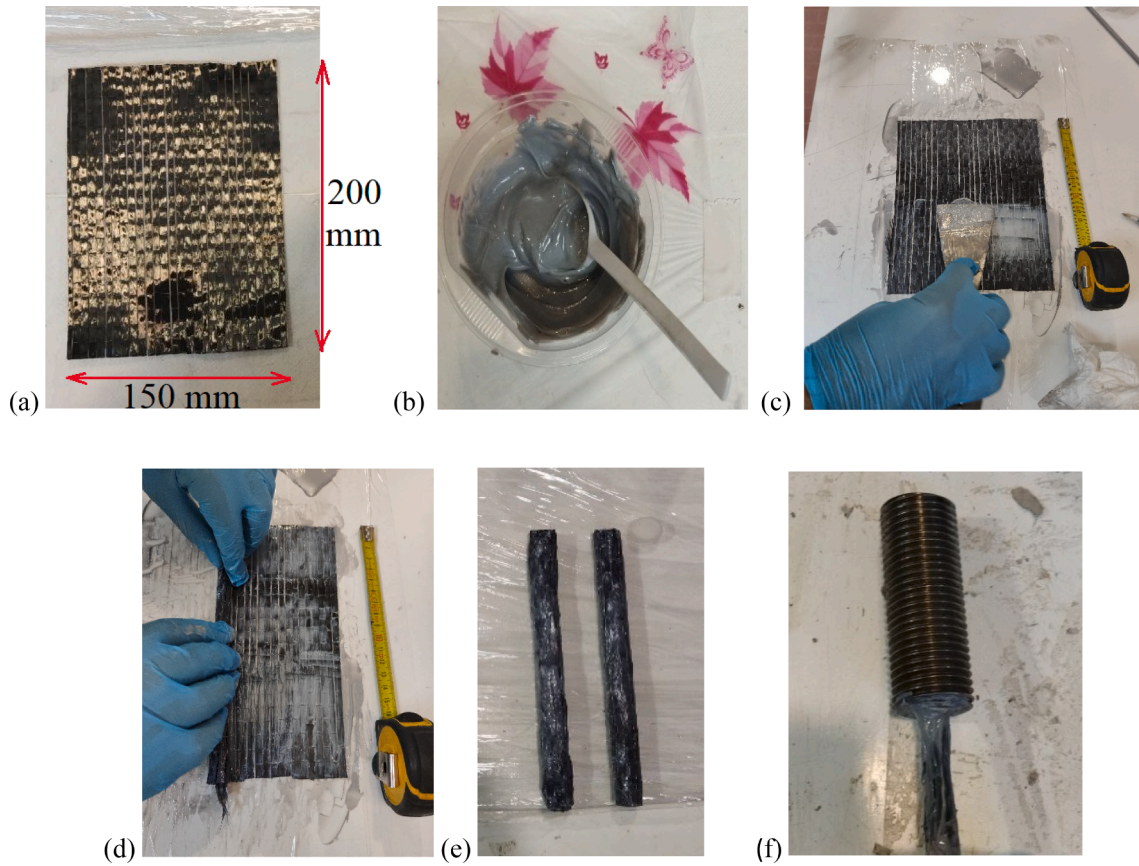


Fig. 3. Preparing FRP anchors: a) cutting a desired size rectangular piece from the FRP sheet, b) preparing epoxy adhesive, c) adding epoxy adhesive to both sides of the FRP piece, d) rolling epoxy-added FRP pieces into a cylinder shape, e) rolled FRP anchor, f) inserting FRP anchor into threaded rod as the sleeve for applying the load.

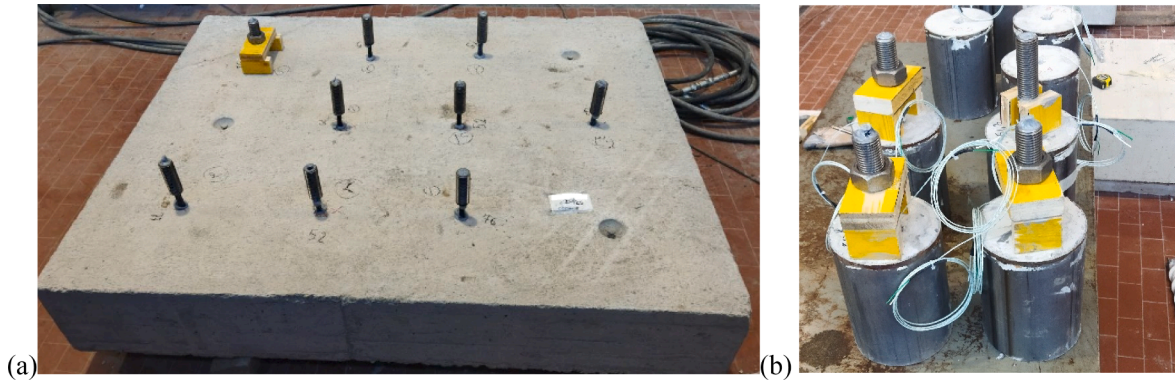


Fig. 4. Inserting FRP anchors into the pre-drilled hole: (a) slab specimen, and (b) cylindrical specimen.

surfaces surrounding the anchors and a piece of concrete detaches with the anchor. The failure mode in this condition is called the concrete cone failure (Fig. 12). When the peak load is reached, the load decreases with a constant slope until the FRP anchor is completely removed from the hole with a concrete cone. On the other hand, at the beginning of loading, the graph slope was the same for all specimens. By continuing the loading procedure, FRP anchors under unconfined conditions showed a lower slope than those in confined conditions. As the loading continues, the samples in the unconfined state take a different path and show less stiffness than the confined samples.

Results of the tests are summarized in Table 5 which reports the observed failure mode and the tensile capacity of FRP anchors, where:

- $F_{u,exp}$ is the pullout capacity of FRP anchor that has been experimentally determined, as a result of a specific failure mode;
 - $F_{ua,exp}$ is mean value of experimental pullout capacities of FRP anchors with the same conditions;
 - τ_u is the average bond strength along the FRP anchor, evaluated as:
- $$\tau_u = F_{u,exp} / (\pi \cdot d_a \cdot h_{ef}) \quad (1)$$
- τ_{ua} is mean value of bond strength of FRP anchors with the same conditions and being d_a the anchor diameter and h_{ef} is the anchor embedment depth.
 - *Std. Dev.* is the standard deviation of the experimental results.

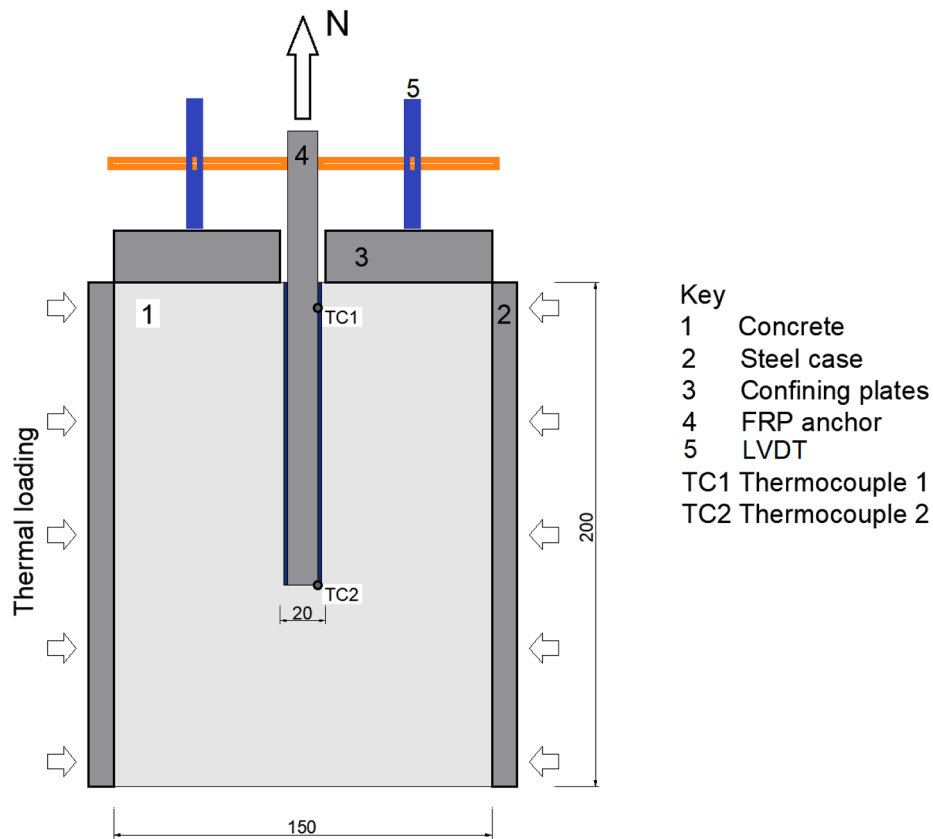


Fig. 5. Geometry of specimen for testing at elevated temperature.

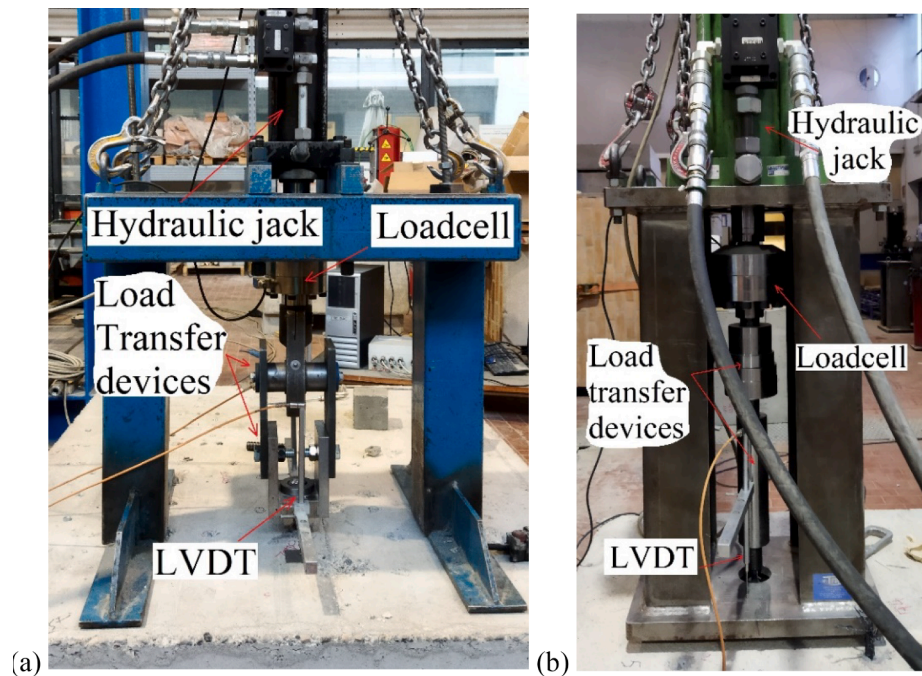


Fig. 6. Test setups for the pullout test on the concrete slab: a) unconfined conditions, b) confined conditions.

As can be seen in Fig. 10, by increasing the embedment length of the FRP anchor from 50 mm to 75 mm in both confined and unconfined conditions, the pull-out capacity of the FRP anchor increases by 54% and 68%, respectively. At the same embedment length, by switching to confined conditions, the pull-out capacity of FRP anchors increases by

65% and 51% for 50 mm and 75 mm embedment, respectively. When varying the length in confined conditions, the average bond strength does not significantly change. However, in unconfined conditions, by increasing the embedment length, the average bond strength increases. This is due to a change in failure modes: from concrete cone failure in

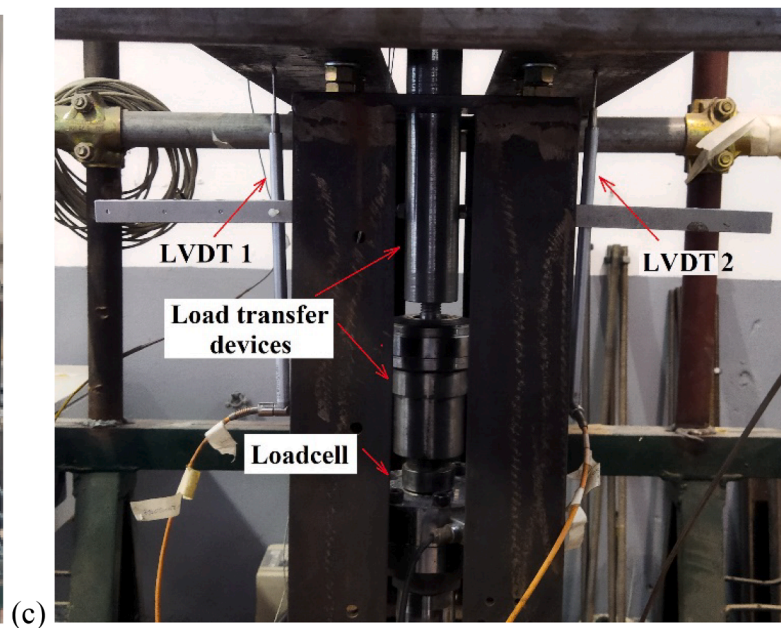
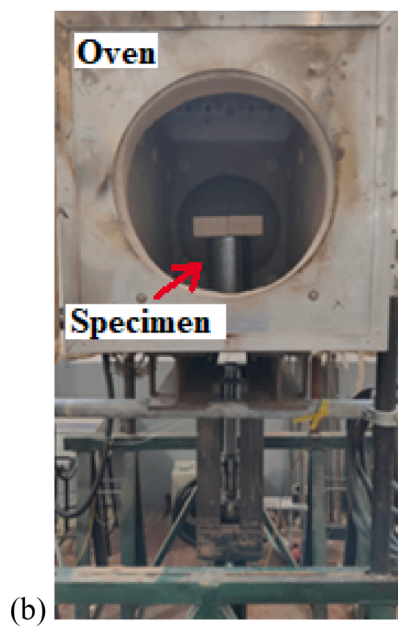
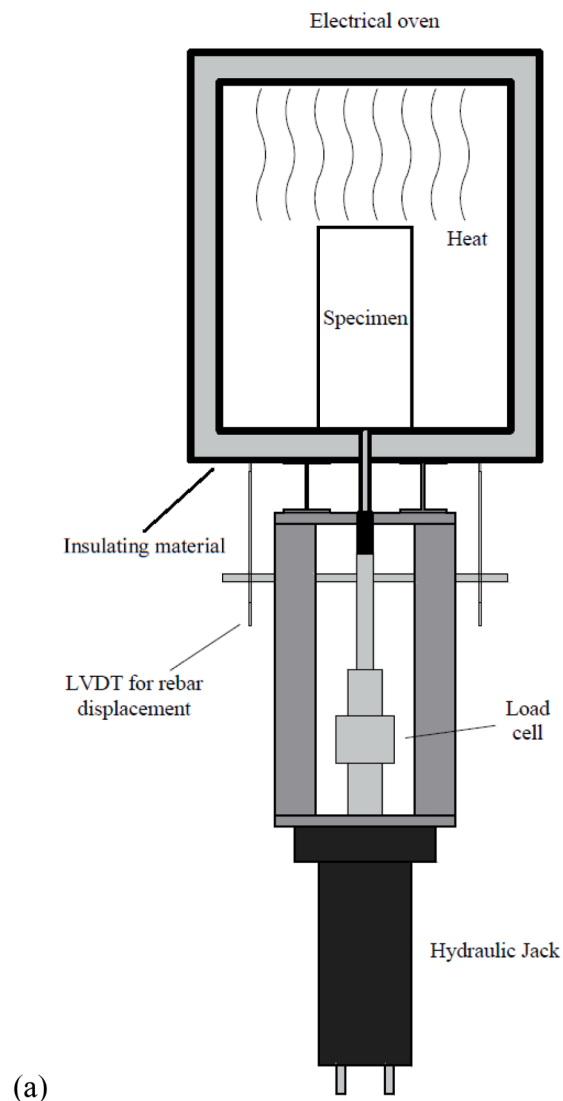


Fig. 7. Test setups for pull-out test on cylindrical concrete specimen under elevated temperature conditions: (a) schematic view of test setup, (b) real test setup, and (c) location of LVDTs.

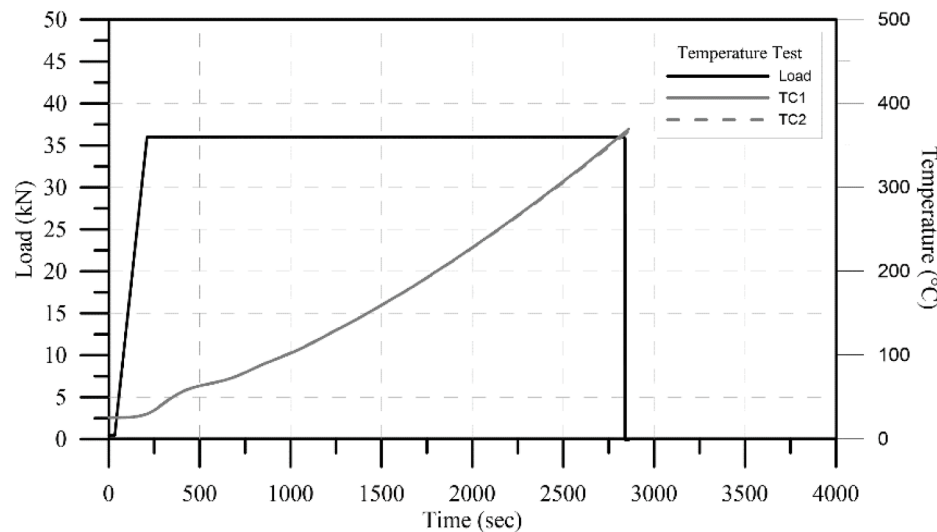


Fig. 8. Load and temperature trends in the specimen with load ratio of 80% during testing in oven.

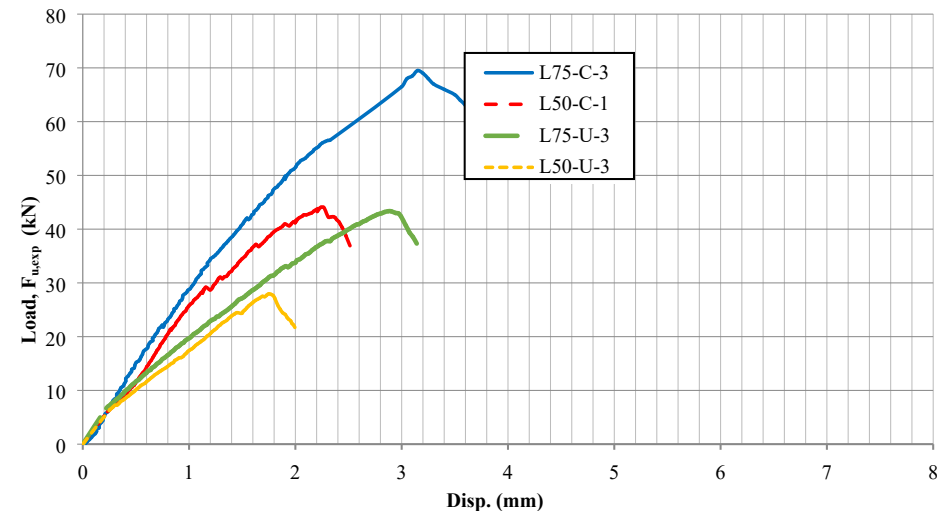


Fig. 9. Load–displacement diagram of FRP anchors under confined and unconfined conditions at ambient temperature.

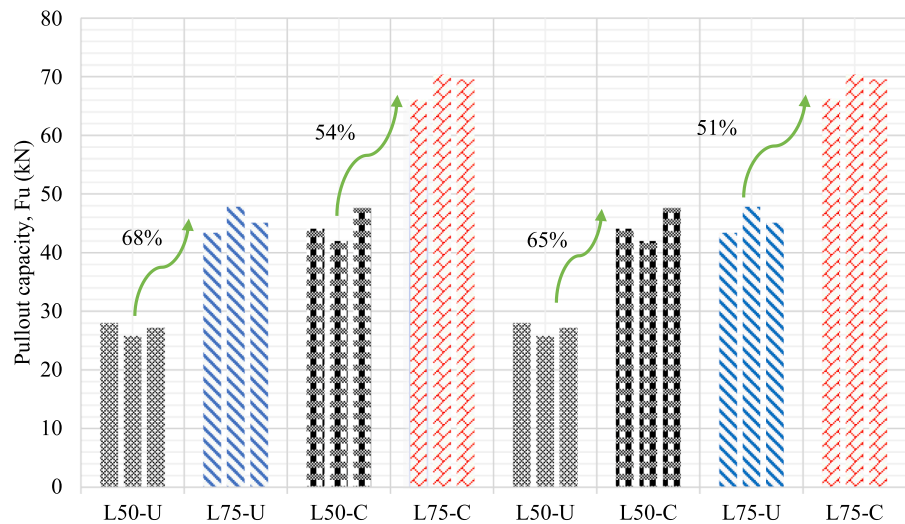


Fig. 10. Comparison of pull-out capacity of FRP anchors at ambient temperature with different embedment lengths and boundary conditions.

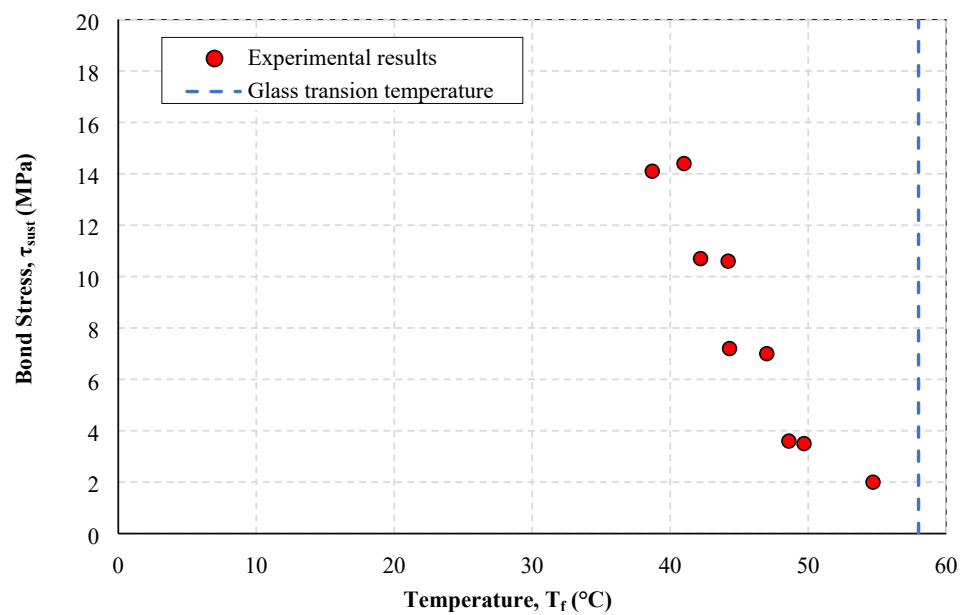


Fig. 11. Bond strength versus temperature for different load ratios of initially applied load.



Fig. 12. Concrete cone failure: (a) top view, (b) lateral view, and (c) comparison concrete cone area of FRP anchors with different embedment lengths.

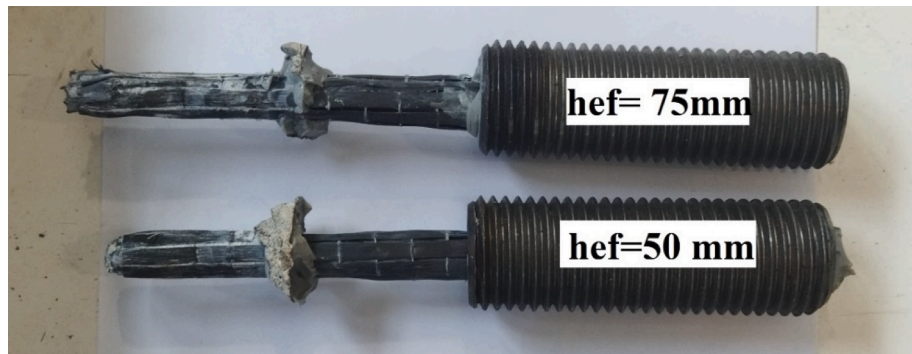


Fig. 13. Pull-out failure (bond failure between FRP anchor and adhesive).

Table 5

Results summary (ambient temperature).

No.	Specimen label	$F_{u, \text{exp}}$ (kN)	$F_{ua, \text{exp}}$ (kN)	Std. Dev. of $F_{u, \text{exp}}$	τ_u (MPa)	τ_{ua} (MPa)	Std. Dev. of τ_u	Failure mode
1	L50-U-1	28.0	27.0	0.9	10.3	10.1	0.2	Concrete cone failure
2	L50-U-2	25.8			9.9			Concrete cone failure
3	L50-U-3	27.2			10.0			Concrete cone failure
4	L75-U-1	43.4	45.4	1.8	11.5	11.9	0.3	Concrete cone failure
5	L75-U-2	47.8			12.3			Concrete cone failure
6	L75-U-3	45.1			11.8			Concrete cone failure
7	L50-C-1	44.1	44.6	2.3	17.2	17.4	0.6	Pullout failure
8	L50-C-2	42.0			16.7			Pullout failure
9	L50-C-3	47.6			18.2			Pullout failure
10	L75-C-1	66.2	68.7	1.8	17.5	18.0	0.4	Pullout failure
11	L75-C-2	70.4			18.2			Pullout failure
12	L75-C-3	69.5			18.5			Pullout failure

unconfined conditions (Fig. 12) into a pure pullout failure mode (Fig. 13), which is accompanied by an increase in FRP anchor strength.

3.1.2. Elevated temperature

Pullout tests at elevated temperatures account for different load ratios of the initially applied load (10%, 20%, 40%, 60%, and 80%) of the mean load detected in the corresponding test series at ambient temperature. A summary of the results of tests conducted at elevated temperatures is provided in Table 6, which, in addition to the quantities previously defined, provides the following information:

- N_{sust} denotes the load applied prior to temperature increase and τ_{sust} refers to the corresponding mean bond stress (calculated according to Eq. (1) by substituting $F_{u, \text{exp}}$ with N_{sust});
- t_1 denotes the test duration;
- δ_f refers to the displacement at failure;
- T_f indicates the temperature at failure.

The temperature T_f is calculated as a weighted average of the temperature $TC1$ (the temperature at the thermocouple near the concrete

surface) and $TC2$ (the temperature at the thermocouple installed end of FRP anchor) recorded at the two thermocouples location along the FRP anchor according to EAD 330087–01-0601 [81] as:

$$T_f = \frac{1}{3}TC1 + \frac{2}{3}TC2 \quad (2)$$

Furthermore, the temperature T_f was identified as the temperature corresponding to the slip of the FRP anchor at the loaded end at failure.

Fig. 11 reports the computed bond stress as a function of the weighted average temperature. As can be seen, the load the anchor is able to sustain decreases with temperature as this approaches the glass transition temperature (the temperature at which an adhesive changes from a hard/glassy state to a soft/leathery state) for the specific bonding agent (as declared by the manufacturer). Moreover, the pull-out of the FRP anchor is associated with the drop in applied load, which is related to the corresponding failure temperature. It is worth noticing that most of the previous research studies on the fire resistance of FRP-strengthened members assumed a perfect bond up to glass transition temperature and a complete debonding thereafter [69]. However, the experiments in the current study showed that bond failure of loaded FRP

Table 6

Result summary (elevated temperatures).

No.	Specimen label	N_{sust} (kN)	h_{ef} (mm)	TC1 (°C)	TC2 (°C)	T_{oven} (°C)	T_f (°C)	τ_{sust} (MPa)	δ_f (mm)	t_1 (sec)
13	L50-R10%-1	5	49.5	60.8	51.7	455.5	54.7	2.0	11.2	3480
14	L50-R20%-1	9	50.6	53.8	47.7	448.4	49.7	3.5	14.8	3612
15	L50-R20%-2	9	50.3	52.9	46.4	452.7	48.6	3.6	11.2	3434
16	L50-R40%-1	18	50.1	47.8	42.5	403.7	44.3	7.2	12.6	3031
17	L50-R40%-2	18	51.3	50.3	45.4	405.9	47.0	7.0	12.9	2768
18	L50-R60%-1	27	50.2	45.4	40.6	405.6	42.2	10.7	11.4	3064
19	L50-R60%-2	27	50.5	45.5	43.6	402.7	44.2	10.6	11.9	3038
20	L50-R80%-1	36	50.8	40.4	37.8	375.2	38.7	14.1	13.3	2981
21	L50-R80%-2	36	49.8	43.4	40.1	364.2	41.2	14.4	14.7	2839

anchor can occur before the adhesive reaches its glass transition temperature. As a general remark, according to Table 6, it can also be noticed that by increasing the constant load, the minimum time to failure decreases.

3.2. Failure modes

Conventional failure modes of the FRP anchors that have been observed in previous research are fiber rupture failure, fan-to-sheet debonding failure, and pull-out failure including concrete-cone failure, bond failure, or combined concrete cone-bond failure [32]. Among the failure modes that exist in the literature, pull-out of FRP anchors has been observed in the current study. The pull-out of the FRP anchor could occur in the form of either concrete pull-out or combined cone-bond pull-out. Concrete cone failure mode is characterized by the pull-out of a concrete cone with an adhesive core. On the other hand, in bond failure, the bond between the adhesive core and concrete is not enough. So, the anchor singly pulls out from the hole. This failure mode occurs for low strength values of the bonding agent at ambient temperature or for any strength of adhesive under the elevated temperature.

3.2.1. Ambient temperature

Typical pictures of FRP anchors at failure are shown in Figs. 12 and 13 pertaining to failure modes reported in Tables 3 and 4. It has been observed that under unconfined testing conditions, a concrete cone was always observed starting from the tip of the anchor (Fig. 12). In fact, the cone depth was equal to the embedment length. At confined conditions, where a bearing pressure was applied to the concrete surface, the FRP anchor was completely pulled out from the surrounding concrete, with a failure surface at the anchor/bonding agent interface (Fig. 13). The failure here was caused by a loss of bond between the lateral face of FRP anchor and adhesive.

3.2.2. Elevated temperature

As shown in Fig. 14, during the test that was conducted at elevated temperatures, all specimens failed due to the loss of the bond between the adhesive in the lateral surface of the FRP anchor and the concrete substrate in the hole.

4. Prediction of anchor capacity and design models

In this section, to predict the capacity of FRP anchors with specific failure modes, the experimental results are compared to the models proposed for adhesive metallic anchors. Models for the evaluation of pull-out strength were primarily developed for the metallic anchors. However, some of them can be used with proper precision for FRP anchors which is presented in section 4.1. It is worth mentioning that some failure modes such as fan-to-sheet debonding or fiber rupture are exclusive failure modes for FRP anchors that need to be examined under a special approach. On the other hand, to investigate the behavior of bonded anchors under elevated temperatures, a bond stress–temperature model should be constructed by plotting bond stress versus failure temperature curves. An investigation of this issue has been carried out in section 4.2, and an equation for FRP anchor bond resistance at high temperatures has been proposed.

4.1. Ambient temperature

To develop the concrete cone models, Concrete cone capacity can be estimated according to the CCD method [82], which was originally developed for post-installed anchors. Using this approach, an angle of approximately 35° (for square projected area) or 45° (for circular projected area) is assumed between the tip of the anchor and the concrete surface, and the capacity is assumed to be directly related to the cone surface and concrete tensile strength, by also accounting for size effects. It has been reported in studies by Fuchs et al. [82], CEB [83], ACI 349-01 [84], ACI 318 M-05 [85], and Eligehausen et al. [86] that a square area should be used, while others have recommended circular areas like Cook [87], ACI 349-85 [88], CEB [83], and Eligehausen et al. [89]. A comparison of the performance of different models at ambient temperature, based on the concrete cone failure mode, can be found in Fig. 15. As it can be seen in Fig. 15, the concrete capacity design (CCD) method proposed by Fuchs et al. [82] is found to be better suited for fitting the experimental data within the limitations generated by the characteristics of the tested FRP anchors. The ultimate capacity of FRP anchor in unconfined conditions with the concrete cone failure mode according to [82] can be taken as:

$$F_{\text{cone, Fuchs et al.}} = 13.5 h_{ef}^{1.5} \sqrt{f_{c,cube}} \quad (3)$$



Fig. 14. Pull-out failure under elevated temperature (bond failure between adhesive and concrete surface).

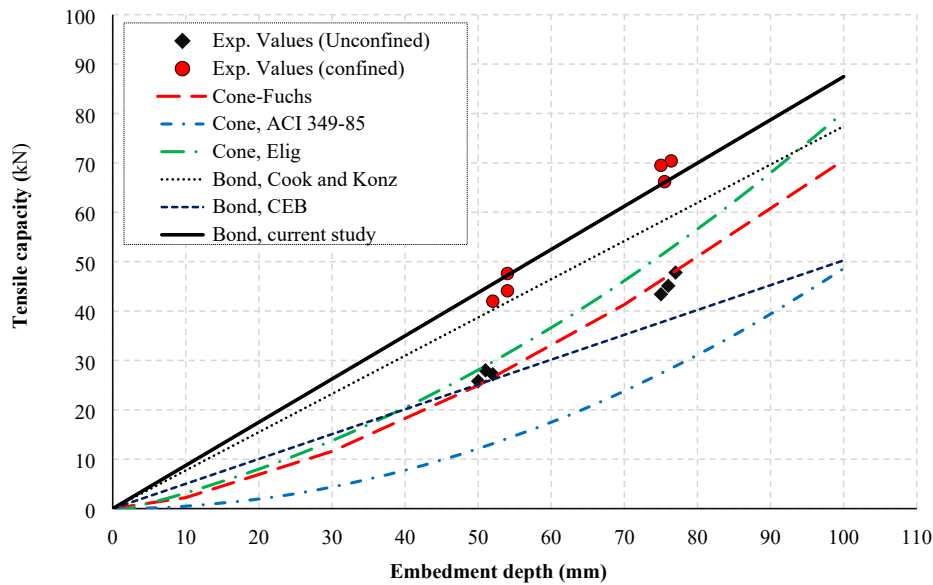


Fig. 15. Anchor capacity as a function of embedment depth, differentiated by various testing conditions, and predictions models according to CCD and UBM approaches.

According to Eq. (4), ACI 349-85 [84] model greatly underestimates the pullout capacity of FRP anchors and is more conservative than Eq. (3), while the model of Eligehausen et al. [86] based on Eq. (5) overestimates the concrete cone strength.

$$F_{cone, ACI 349-85} = 0.93 h_{ef}^2 \sqrt{f_{c,cube}} \quad (4)$$

$$F_{cone, Elig et al.} = 15.3 h_{ef}^{1.5} \sqrt{f_{c,cube}} \quad (5)$$

In the bonded interface resistance models, the failure surface is assumed to be adjacent to the embedded portion of the anchor at the anchor-to-concrete interface. In bond models, the most important parameters are the failure surface area and the bond stress (τ) of the anchor-to-concrete interface which is generally assumed to be uniform. Therefore, under confined conditions applied, being concrete cone prevented, a uniform bond model (UBM) can be adopted assuming an embedment-independent bond strength value. A model based on this approach was proposed by Cook [87] and CEB [83] using Eq. (6).

$$F_{bond, Cook and CEB} = \tau \cdot \pi \cdot d_h \cdot h_{ef} \quad (6)$$

In this equation, the coefficient τ changes from 6.3 to 14.6 MPa (average of 10.4) and 8 MPa according to Cook [87] and CEB [83], respectively. This equation underestimates the capacity of FRP anchor with bond failure. Meanwhile, Cook and Kunz [90] and Eligehausen et al. [91] recommend Eq. (7) to predict the anchor strength. This equation is similar to Eq. (6), with the difference that d_h is replaced by d_a . Moreover, based on information presented by Cook and Konz [90], the value of τ is 15.4 MPa and 12 MPa according to the tested value and design purpose, respectively.

$$F_{bond, Cook and Konz} = \tau \cdot \pi \cdot d_a \cdot h_{ef} \quad (7)$$

According to the comparison of the experimental results with the relations presented in the mentioned models, it can be concluded that the parameter of bond stress (τ) should be determined based on the tested values obtained from the related research, which in this study included FRP anchors with a diameter of 16 mm and embedment lengths of 50 or 75 mm. Thus, formulation for calculating the bond strength of FRP anchors followed by bond failure according to current test data, returning a mean bond strength (τ_{avg}) value equal to 17.7 MPa. In conclusion, Fig. 15 reports the experimentally determined anchor capacities as a function of the anchor embedment depths, comparing them

with the predictions according to Eqs. (3)–(8). As it can be interpreted from Eqs. (3)–(8), in the anchors followed by the concrete cone failure, the strength of concrete becomes a decisive parameter, differently from the case when bond failure mode is dominant, where the concrete compressive strength does not have any special influence.

It can be noticed how the CCD is perfectly suitable to estimate anchor capacity in unconfined cases, which is significantly relevant for real applications, where such bond conditions are present. However, under such conditions, only a lower bound estimation of anchor bond strength is possible. Consequently, assessing the bond strength through confined testing proves to be efficient in this direction. The UBM fits the experimental results well, returning a mean bond strength (τ_{avg}) value of 17.7 MPa. It should be noted that this value, obtained under confined conditions, cannot be used directly in design but rather as a reference value when assessing other effects, such as the influence of temperature, creep, and installation conditions. However, if the bond strength obtained under confined conditions is used to predict the capacity associated with a combined pullout/concrete cone failure mode (as in [92]), the results suggest an unconfined-to-confined ratio (referred to as α setup in EAD 330499 [65] qualification procedures) of 0.5.

It is worth mentioning that by using Eq. (1), it is assumed that the stress distribution is the same along the whole length of the dowel, when in reality a non-uniform slip distribution along the dowel implies higher stresses toward the surface of the concrete. In the case of a very long dowel, after having reached the transfer length, the bottom would not experience any stress. At the current status of investigation, it is suggested to adopt such approach only for short lengths in view of future extensions. To such regard, it is noticed that the general framework of EN 1992-4 [93] for bonded anchors, which also inspires the adopted approach, assumes that a uniform bond model adopting bond strength evaluated on the basis of confined tests (carried out on the samples with a ratio h_{ef}/d_a not larger than 10 that are considered as anchors with a short dowel length according to [65]– as in the current work) can be applied in the design range $h_{ef}/d_a = 4$ –20.

4.2. Elevated temperature

A bond stress–temperature model was constructed by plotting bond stress versus failure temperature curves for FRP anchors evaluated at elevated temperatures and confined conditions and estimating coefficients by inverse analysis. As can be seen in Fig. 16, an exponential

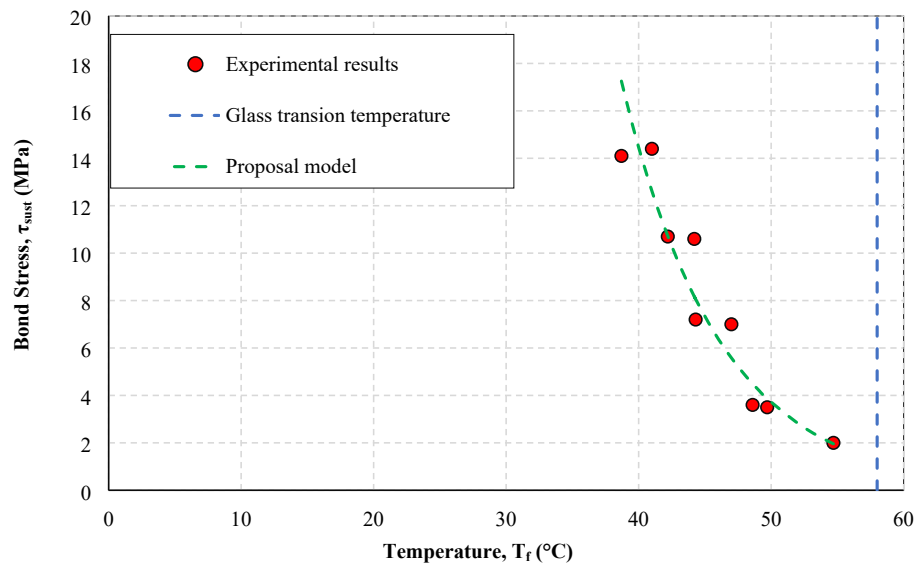


Fig. 16. Bond strength versus temperature estimated by proposed equation and experimental result.

curve is able to properly fit the experimental result (with an R^2 value equal to 0.935) while after the glass transition temperature too few results exist to provide a reasonable prediction. Under elevated temperatures, an exponential curve is recommended in EAD 330087-01-0601 [81] for post-installed rebar connections with mortar under elevated temperature. According to the experimental findings, FRP anchors with a depth of 50 mm, even when loaded at 20% of their capacity, have been shown to fail before their adhesive reaches its glass transition temperature due to an increase in ambient temperature and subsequent temperature within the concrete.

It is worth to discuss that the obtained curve represents a decay of the entire anchor capacity, but rather of its local bond strength. This decay was experimentally evaluated on specimens whose bond length is short enough to assume that the average stress along the dowel at failure may be associated with the local bond strength at the observed temperature of failure. Such result will successively represent the input for an assessment of the effect of temperatures on an anchor which may be evaluated through a global analysis of a structure/member under a fire event [94]. This analysis aims to determine the effects of temperature distribution along the anchor for a specific fire duration. On the other hand, the load transfer capacity can then be evaluated by integrating the bond stress along the bond length, as originally proposed [95] and successively validated for anchorages in concrete by Pinoteau et al. [96]. This may imply that only the inner layers, where temperature penetration is not such to induce a too significant decay of the bond strength, will effectively contribute to the load carrying capacity of the anchor.

However, the full applicability of such procedure to FRP anchors is outside of the scope of the present paper and will require specific analyses and validation in the future.

5. Conclusions

The current study was conducted to assess the behavior of FRP anchors with dowels ranging from 50 to 75 mm under tensile loading, both under confined and unconfined conditions, at ambient room and elevated temperatures. To achieve this goal, a variety of testing setups were used, and the research procedure was divided into two steps to evaluate the bond behavior of FRP anchors installed in concrete specimens. The first step involved investigating the effect of confinement and different embedment depths on the strength capacity and failure modes of FRP anchors at ambient temperature. Therefore, three concrete slabs

were cast, and after the installation of the FRP anchors, they were tested in confined and unconfined conditions. Moreover, experimental results were compared with models for predicting the bond strength of post-installed fasteners. In the second step, by conducting eleven pullout tests on the cylindrical specimens, the effects of confinement and elevated temperature on the bond properties of FRP anchors were studied. Furthermore, a model was proposed to determine the relationship between the bond strength of FRP anchors and temperature. Based on the results of the experimental study, which included pull-out capacity and displacement at failure, different failure modes, and control of temperature distribution along the FRP anchor, as well as a comparison with existing models available in the literature, the following conclusions can be drawn:

1. Longer embedment lengths increase FRP anchor capacity. When anchors are embedded deeper, the peak load increases. This increase is predicted by a 1.5 exponent power law function of embedment depth, which is strictly related to the formation of a concrete cone starting from the tip of the anchor at failure.
2. When a bearing pressure is applied, and so the concrete cone is prevented in favor of a bond failure at the anchor interface, the anchor capacity significantly increases. Moreover, the anchor capacity can be well estimated by adopting a uniform bond model (UBM) if the dowel length of the FRP anchor is short with respect to the anchor diameter. Additional investigations would be required to extend the validity of such approach also to higher embedment depths, as in the approach of EN 1992-4 [93].
3. The capacity of FRP anchors can then be estimated by confined pullout tests in concrete by adopting the same approach currently used for post-installed fasteners in concrete. First, the proposed relation of different equations was assessed with respect to its goodness of fit using certain statistical indices, which revealed a good agreement between estimated and measured values.
4. The ratio between constant load and short term load an FRP anchor can sustain rapidly decreases when approaching the glass transition temperature of the bonding agent.
5. Testing under confined conditions where a constant load is applied before temperature increase can be adopted to obtain a bond strength vs. temperature relationship. An exponential curve properly describes the decay of the bond strength as function of the temperature at the anchor/concrete interface for an FRP anchor.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

The tests were conducted at the Material Testing Laboratory (LPMSC) of Milan Polytechnic, which is thanked for the financial support provided. The authors would like to express their gratitude to Daniele Spinelli and Michele Dezio for their support during the experimental activities. During the visiting research period of the first author at the Polytechnic of Milan, financial support was offered by the Iranian Ministry of Science, Research and Technology (MSRT), and the Isfahan University of Technology (IUT) which is admirably acknowledged.

References

- Hollaway L. A review of the present and future utilisation of FRP composites in the civil infrastructure with reference to their important in-service properties. *Constr Build Mater* 2010;24(12):2419–45.
- Al-Mahaidi R, Kalfat R. Rehabilitation of concrete structures with fiber-reinforced polymer. Butterworth-Heinemann; 2018.
- Naser MZ, Hawileh RA, Abdalla JA. Fiber-reinforced polymer composites in strengthening reinforced concrete structures: a critical review. *Eng Struct* 2019;198:109542.
- Mertoglu C, Anil O, Durucan C. Bond slip behavior of anchored CFRP strips on concrete surfaces. *Constr Build Mater* 2016;123:553–64.
- Teng J, Smith ST, Yao J, Chen J-F. Intermediate crack-induced debonding in RC beams and slabs. *Constr Build Mater* 2003;17(6–7):447–62.
- Aram MR, Czaderski C, Motavalli M. Debonding failure modes of flexural FRP-strengthened RC beams. *Compos B Eng* 2008;39(5):826–41.
- Teng J, Chen J-F, Yu T. FRP-strengthened RC structures. West Sussex, Eng: John Wiley Sons LTD; 2002.
- Mostofinejad D, Mahmoodabadi E. Grooving as alternative method of surface preparation to postpone debonding of FRP laminates in concrete beams. *J Compos Constr* 2010;14(6):804–11.
- Sanginabadi K, Yazdani A, Mostofinejad D, Czaderski C. RC members externally strengthened with FRP composites by grooving methods including EBROG and EBRIG: a state-of-the-art review. *Constr Build Mater* 2022;324:126662.
- Sanginabadi K, Yazdani A, Mostofinejad D, Czaderski C. Bond behavior of FRP composites attached to concrete using EBROG method: a state-of-the-art review. *Compos Struct* 2022;299:116060.
- Ghahsareh FM, Mostofinejad D, Batebi S. Effects of different groove classes used in externally bonded reinforcement on grooves joints on carbon fiber-reinforced polymer-to-concrete bond behavior. *ACI Struct J* 2022;119(4).
- Ghahsareh FM, Mostofinejad D. Groove classification in EBROG FRP-to-concrete joints. *Constr Build Mater* 2021;275:122169.
- Foraboschi P. Strengthening of reinforced concrete beams subjected to concentrated loads using externally bonded fiber composite materials. *Materials* 2022;15(6):2328.
- Kalfat R, Al-Mahaidi R, Smith ST. Anchorage devices used to improve the performance of reinforced concrete beams retrofitted with FRP composites: State-of-the-art review. *J Compos Constr* 2013;17(1):14–33.
- Kim I. Use of CFRP to provide continuity in existing reinforced concrete members subjected to extreme loads. department of civil, environmental and architectural engineering. Austin, Texas, The University of Texas at Austin, Ph. D Dissertation; 2008.
- del Rey Castillo E, Dizhur D, Griffith M, Ingham J. Strengthening RC structures using FRP spike anchors in combination with EBR systems. *Compos Struct* 2019;209:668–85.
- Foraboschi P. Layered plate with discontinuous connection: exact mathematical model. *Compos B Eng* 2013;47:365–78.
- Kim Y, Ghannoum WM, Jirsa JO. Shear behavior of full-scale reinforced concrete T-beams strengthened with CFRP strips and anchors. *Constr Build Mater* 2015;94:1–9.
- Zaki MA, Rasheed HA, Alkhrdaji T. Performance of CFRP-strengthened concrete beams fastened with distributed CFRP dowel and fiber anchors. *Compos B Eng* 2019;176:107117.
- Hosseini SM, Mostofinejad D. Seismic performance of RC short columns retrofitted with a novel system in shear and flexure using CFRP composites. *J Compos Constr* 2021;25(5):04021039.
- del Rey Castillo E, Griffith M, Ingham J. Seismic behavior of RC columns flexurally strengthened with FRP sheets and FRP anchors. *Compos Struct* 2018;203:382–95.
- Akhlaghi A, Mostofinejad D. Experimental and analytical assessment of different anchorage systems used for CFRP flexurally retrofitted exterior RC beam-column connections. *Structures* 2020;28:881–93.
- Ila E, Mostofinejad D, Moghaddas A. Cyclic behavior of strong beam-weak column joints strengthened with different configurations of CFRP sheets. *Arch Civil Mech Eng* 2020;20(2):1–26.
- Mostofinejad D, Hajrasouliha M. 3D beam-column corner joints retrofitted with X-shaped FRP sheets attached via the EBROG technique. *Eng Struct* 2019;183:987–98.
- Meisami MH, Mostofinejad D, Nakamura H. Strengthening of flat slabs with FRP fan for punching shear. *Compos Struct* 2015;119:305–14.
- Smith ST, Zhang H, Wang Z. Influence of FRP anchors on the strength and ductility of FRP-strengthened RC slabs. *Constr Build Mater* 2013;49:998–1012.
- Altin S, Anil O, Koprman Y, Kara ME. Hysteretic behavior of RC shear walls strengthened with CFRP strips. *Compos B Eng* 2013;44(1):321–9.
- Abdulsalam B, Ali AH, ElSafty A, Elshafey N. Behavior of GFRP strengthening masonry walls using glass fiber composite anchors. *Structures* 2021;29:1352–61.
- Fagone M, Rotunno T, Bertolesi E, Grande E, Milani G. Experimental study on the local behavior of CFRP anchor spikes fixed to masonry substrates. *Trans Tech Publ Key Eng Mater* 2022:361–8.
- Zhang H, Smith ST, Gravina RJ, Wang Z. Modelling of FRP-concrete bonded interfaces containing FRP anchors. *Constr Build Mater* 2017;139:394–402.
- Muciaccia G, Khorasani M, Mostofinejad D. Effect of confinement on the behavior of FRP straight anchors, Bond in Concrete 2022. Stuttgart, Germany; 2022.
- Muciaccia G, Khorasani M, Mostofinejad D. Effect of different parameters on the performance of FRP anchors in combination with EBR-FRP strengthening systems: a review. *Constr Build Mater* 2022;354:129181.
- Ozbakkaloglu T, Saatcioglu M. Tensile behavior of FRP anchors in concrete. *J Compos Constr* 2009;13(2):82–92.
- del Rey Castillo E, Kanitkar R. Effect of FRP spike anchor installation quality and concrete repair on the seismic behavior of FRP-strengthened RC columns. *J Compos Constr* 2021;25(1):04020085.
- ACI440.2R. Guide for the design and construction of externally bonded FRP Systems. American Concrete Institute Farmington Hills, Michigan; 2017.
- JSCE-E541. Japanese Design and construction guidelines for seismic retrofit of existing reinforced concrete buildings. Japan Society of Civil Engineers (JSCE), Tokyo, Japan; 2001.
- Sun W, Liu H, Wang Y, He T. Impacts of configurations on the strength of FRP anchors. *Compos Struct* 2018;194:126–35.
- Shekarchi W, Pudliner D, Alotaibi N, Ghannoum W, Jirsa J. Carbon fiber-reinforced polymer spike anchor design recommendations. *ACI Struct J* 2020;117(6).
- Koutas L, Triantafillou TC. Use of anchors in shear strengthening of reinforced concrete T-beams with FRP. *J Compos Constr* 2013;17(1):101–7.
- Zhang H, Smith ST. Influence of FRP anchor fan configuration and dowel angle on anchoring FRP plates. *Compos B Eng* 2012;43(8):3516–27.
- Ozdemir G, Akyuz U. Tensile capacities of CFRP anchors. *Spec Publ* 2005;230:39–56.
- Orton SL, Jirsa JO, Bayrak O. Design considerations of carbon fiber anchors. *J Compos Constr* 2008;12(6):608–16.
- Kim I. Rehabilitation of poorly detailed RC structures using CFRP materials. University of Texas; 2006.
- Huaco GD. Quality control test for carbon fiber reinforced polymer (CFRP) anchors for rehabilitation. The University of Texas at Austin, Austin, Texas, 2009.
- Fib-Bulletin-90. Externally bonded FRP reinforcement for RC structures, fib Task Group 5.1, Lausanne, Switzerland; 2019.
- Quinn KT. Shear strengthening of reinforced concrete beams with carbon fiber reinforced polymer (CFRP) and improved anchor details; 2009.
- Pham L. Development of a quality control test for carbon fiber reinforced polymer anchors. Master's thesis, The University of Texas at Austin, Austin, TX; 2009.
- Pudliner D, Ghannoum WM, Jirsa JO. Influence of anchor size on anchored CFRP systems. *J Compos Constr* 2019;23(5):04019033.
- Kalfat R, Al-Mahaidi R. Improvement of FRP-to-concrete bond performance using bidirectional fiber patch anchors combined with FRP spike anchors. *Compos Struct* 2016;155:89–98.
- Singh A, del Rey Castillo E, Ingham J. FRP-to-FRP bond characterization and force-based bond length model. *Compos Struct* 2019;210:724–34.
- Sun W, Liu S, Zhang C. An effective improvement for enhancing the strength and feasibility of FRP spike anchors. *Compos Struct* 2020;247:112449.
- Pudliner DK. Design considerations based on size effects of anchored carbon fiber reinforced polymer (CFRP) systems. USA: The University of Texas at Austin; 2016.
- Kobayashi K, Fujil S, Yabe Y, Tsukagoshi H, Sugiyama T. Advanced wrapping system with CF-anchor-Stress transfer mechanism of CF-anchor, FRPRCS-5: Fibre-reinforced plastics for reinforced concrete structures Volume 1: Proceedings of the fifth international conference on fibre-reinforced plastics for reinforced concrete structures, Cambridge, UK, 16–18 July 2001, Thomas Telford Publishing; 2001. p. 379–88.
- Niemitz CW, James R, Breña SF. Experimental behavior of carbon fiber-reinforced polymer (CFRP) sheets attached to concrete surfaces using CFRP anchors. *J Compos Constr* 2010;14(2):185–94.
- del Rey Castillo E, Griffith M, Ingham J. Straight FRP anchors exhibiting fiber rupture failure mode. *Compos Struct* 2019;207:612–24.
- del Rey Castillo E, Dizhur D, Griffith M, Ingham J. Experimental testing and design model for bent FRP anchors exhibiting fiber rupture failure mode. *Compos Struct* 2019;210:618–27.

- [57] Al-Sammari AT, Breña SF. Strength of carbon fiber-reinforced polymer (CFRP) sheets bonded to concrete with CFRP spike anchors. *ACI Struct J* 2021;118(1).
- [58] Eligehausen R, Mallée R, Silva JF. Anchorage in concrete construction. John Wiley & Sons; 2006.
- [59] Marchisella A, Muciaccia G. Influence of specimen geometry on the response of post-installed anchors subjected to constant load under crack opening and crack cycles. *J Test Eval* 2020;48(5):1–20.
- [60] Mahrenholtz P, Eligehausen R, Hutchinson TC, Hoehler MS. Behavior of post-installed anchors tested by stepwise increasing cyclic load protocols. *ACI Struct J* 2016;113(5):997.
- [61] Faraone G, Hutchinson TC, Piccinin R, Silva JF. Anchor performance in cyclically loaded shear walls. *ACI Struct J* 2022;119(6).
- [62] Razvi SWN, Shaikh M. Effect of confinement on behavior of short concrete column. *Proc Manuf* 2018;20:563–70.
- [63] Micelli F, Cascardi A, Aiello MA. Pre-load effect on CFRP-confinement of concrete columns: experimental and theoretical study. *Crystals* 2021;11(2):177.
- [64] ACI355.4. Qualification of Post-installed Adhesive Anchors in Concrete (ACI 355.4-19) and Commentary: An ACI Standard. American Concrete Institute; 2019.
- [65] EAD330499-EOTA. Bonded Fasteners for Use in Concrete, EOTA Brussels, Belgium; 2019.
- [66] ACI318-19. Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary, ACI Committee 318, Farmington Hills, MI, USA; 2019.
- [67] CEN. Eurocode 4, Design of Composite Steel and Concrete Structures, Part 1.1: General Rules and Rules for Building, BS EN 1994-1-1: 2004; 2003.
- [68] Firmo JP, Correia JR, Bisby LA. Fire behaviour of FRP-strengthened reinforced concrete structural elements: a state-of-the-art review. *Compos B Eng* 2015;80: 198–216.
- [69] Ahmed A, Kodur V. Effect of bond degradation on fire resistance of FRP-strengthened reinforced concrete beams. *Compos B Eng* 2011;42(2):226–37.
- [70] Wu Z, Iwashita K, Yagashiro S, Ishikawa T, Hamaguchi Y. Temperature effect on bonding and debonding behavior between FRP sheets and concrete. *J Soc Mater Sci* 2005;54(5):474–80.
- [71] Muciaccia G, Consiglio A, Rosati G. Creep behavior of bonded anchor under high sustained loading at long term temperature. *High Tech Concrete: Where Technology and Engineering Meet: Proceedings of the 2017 fib Symposium, held in Maastricht, The Netherlands, June 12–14, 2017, Springer; 2018. p. 956–64.*
- [72] Pinoteau N, Pimienta P, Guillet T, Rivillon P, Rémond S. Effect of heating rate on bond failure of rebars into concrete using polymer adhesives to simulate exposure to fire. *Int J Adhes Adhes* 2011;31(8):851–61.
- [73] Muciaccia G, Consiglio A, Rosati G. Behavior and design of post-installed rebar connections under temperature. *Key engineering materials, Trans Tech Publ; 2016. p. 783–90.*
- [74] Muciaccia G, Consiglio AN, Rosati G. Recent developments in design of post-installed rebar connections under temperature. *Connections between Steel and Concrete (ConSC 2017), IWB Stuttgart; 2017. p. 181.*
- [75] Sika G. Product data sheet, SikaWrap-300C, Woven carbon fiber fabric for structural strengthening; 2017. www.gcc.sika.com.
- [76] Sika G. Technical data sheet, Sikadur-330, 2-part thixotropic epoxy adhesive; 2019. www.sika.com.
- [77] Astmd3039, d3039m,. Standard test method for tensile properties of polymer matrix composite materials. West Conshohocken, PA: ASTM International; 2000.
- [78] Eurocode 2, E.C.f. Standardization. Design of concrete structures; 2004.
- [79] EN12390-3. Testing hardened concrete. Compressive strength of test specimens; 2009.
- [80] Muciaccia G, Navarrete DD, Pinoteau N, Mége R. Effects of different test apparatus and heating procedures on the bond properties of post-installed rebar connections under elevated temperatures. *Mater Struct* 2019;52(2):1–11.
- [81] EAD330087-01-0601. Systems for post-installed rebar connections with mortar. European Organization for Technical Assessment, Brussels, Belgium; 2020.
- [82] Fuchs W, Eligehausen R, Breen JE. Concrete capacity design (CCD) approach for fastening to concrete. *Structural Journal* 1995;92(1):73–94.
- [83] CEB. Fastenings to Concrete and Masonry Structures: State of the Art Report. Comité euro-international du béton; 1994.
- [84] ACI349-01. Code requirements for nuclear safety related concrete structures and commentary. American Concrete Institute (ACI), Detroit, USA; 2001.
- [85] ACI318M-05. Building code requirements for structural concrete and commentary. American Concrete Institute (ACI), Detroit, USA; 2005.
- [86] Eligehausen R, Fuchs W, Mayer B. Tragverhalten von Dübelbefestigungen unter Zugbeanspruchung (Bearing behaviour of anchor fastenings under tension). *Betonwerk Fertigteil-Tech* 1987;12:826–32.
- [87] Cook RA. Behavior of chemically bonded anchors. *J Struct Eng* 1993;119(9): 2744–62.
- [88] ACI349-85. Code requirements for nuclear safety related concrete structures. American Concrete Institute (ACI), Detroit, USA; 1985.
- [89] Eligehausen R, Mallée R, Rehm G. Befestigungen mit Verbundankern (Fastenings with bonded anchors). *Betonwerk Fertigteil-Tech* 1984;10:682–92.
- [90] Cook RA, Konz RC. Factors influencing bond strength of adhesive anchors. *Structural Journal* 2001;98(1):76–86.
- [91] Eligehausen R, Cook RA, Appl J. Behavior and design of adhesive bonded anchors. *ACI Struct J* 2006;103(6):822.
- [92] BS-EN1994-1-1. Design of composite steel and concrete structures. General rules and rules for buildings, London, UK; 2004.
- [93] EN1992-4. Eurocode 2: Design of Concrete Structures—Part 4: Design of Fastenings for Use in Concrete. European Committee for Standardization (CEN); EN (2018) 1992-4.
- [94] Muciaccia G. Anchorage and detailing of reinforcement under fire action. *Struct Concr* 2023;24(3):3264–80.
- [95] Pinoteau N, Guillet T, Rémond S, Pimienta P, Mége R. Background on the fire evaluation of post-installed reinforcement bars in concrete. In: *Proceedings of the 3rd international symposium ConSC2017 (connections between steel and concrete)*, Stuttgart, Germany; 2017. p. 27–9.
- [96] Pinoteau N, Rémond S, Pimienta P, Rivillon P, Guillet T. Validation of a method to determine the strength of epoxy-bonded Rebars in concrete with a full-scale ISO 834-1 fire test. In: *The 8th international conference on structures in fire, Shanghai (China); 2014. p. 327–34.*